

HW4_Simulation Section (EE576)

Design and Simulation Problem

Instructor: Dr. Subhanshu Gupta

By: Mohammad Chahardori

ID:11632753

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Simulation Problem:

Convert the design in HW3 to a fully-differential amplifier (only change is the load capacitor).

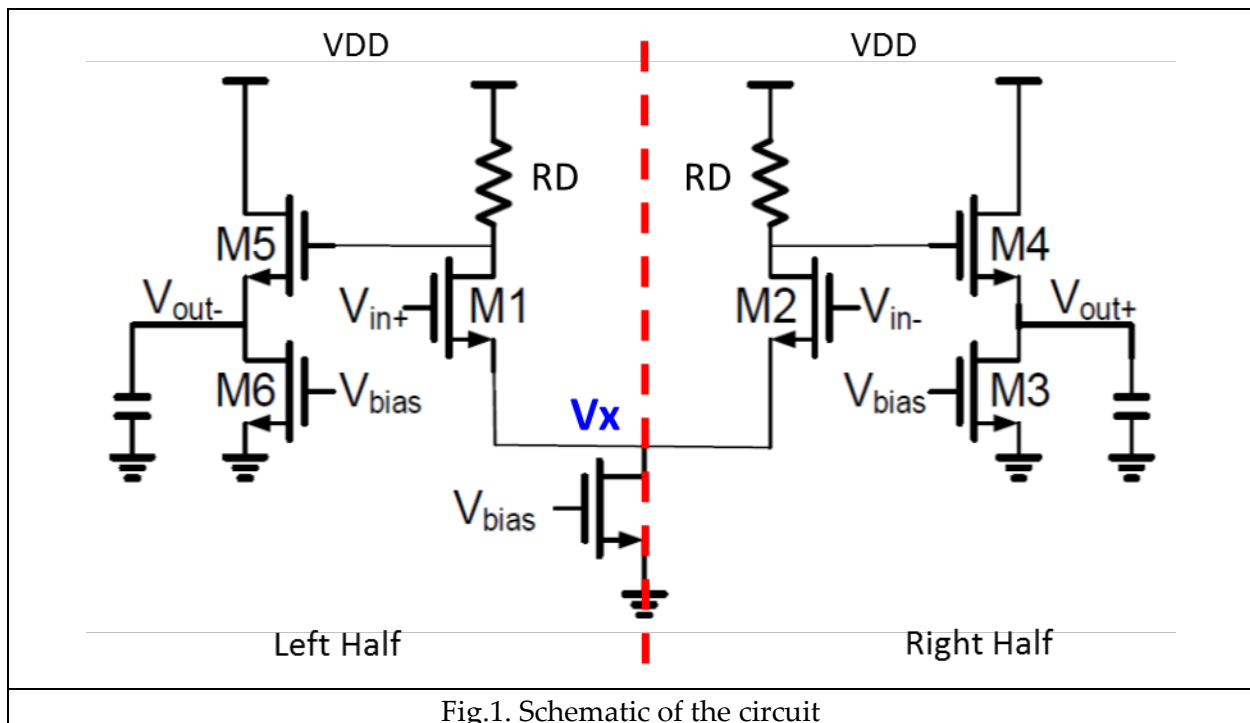
Considering load resistor $R_D=500\Omega$, the capacitor C of 20pF, and $V_{dd}=1.3V$, you need to design a 2- stage fully-differential cascaded common-source amplifier with a source follower driving 20pF.

You are to use an 130nm CMOS process. The design goals for this circuit are as follows:

- 1- Low frequency differential ac voltage gain $> 12\text{dB}$
- 2- Low-frequency common-mode Rejection Ratio (CMRR) $> 80\text{dB}$
- 3- Maximum output swing $> 0.6V_{\text{pk-pk-diff}}$ (peak-to-peak-differential)
- 4- Unity gain bandwidth $> 500\text{MHz}$

Answer to the Simulation Problem:

In fig.1 the basic schematic of the circuit has been shown.



According to the required specification of the amplifier, the minimum low frequency differential gain of the amplifier is 12 dB which is about 4. To satisfy this gain after final circuit design, we consider 20% margin for this gain and assume the overall gain is 5. On the other hand, according to the differential half circuit structure (Right half of the circuit in fig.1) and considering the node Vx (sources of M1 and M2) is virtual ground in differential mode because it is placed on symmetry line, the amplifier differential gain in low frequency is:

$$Amp_diff_Gain \geq 5$$

also

$$Amp_diff_Gain = Gain_1 \times Gain_2 = -g_{m2} \cdot (R_D \parallel r_{o2}) \times \frac{g_{m4}}{g_{m4} + g_{mb4} + g_{o3} + g_{o4}} \approx -g_{m2} \cdot R_D \times \frac{g_{m4}}{g_{m4} + g_{mb4}}$$

and since

$$\frac{g_{m4}}{g_{m4} + g_{mb4}} \approx 0.7$$

so

$$Gain_1 = g_{m2} \cdot R_D \geq 7$$

for

$$R_D = 500\Omega$$

→

$$g_{m2} \geq 14^{mS} \rightarrow \text{with 25\% margin} \rightarrow g_{m2} = 18^{mS}$$

because of symetry in differential pair:

$$g_{m1} = g_{m2} = 18^{mS}$$

(1)

The other low frequency specification of the circuit is the required CMRR > 80dB. According to the CMRR definition we have:

$$CMRR = \frac{A_{V_Diff}}{A_{V_CM}}$$

→

$$CMRR(dB) = A_{V_Diff}(dB) - A_{V_CM}(dB)$$

(2)

Also based on the problem specification, we have:

$$CMRR = 80 \text{ dB}$$

→

$$A_{V_Diff}(dB) - A_{V_CM}(dB) = 80 \text{ dB}$$

→

$$12 \text{ dB} - A_{V_CM}(dB) = 80 \text{ dB} \quad (3)$$

→

$$A_{V_CM}(dB) = -68 \text{ dB}$$

→

$$A_{V_CM} = 0.0004$$

Now we need to calculate the common mode gain of the amplifier. Because of the symmetry of the circuit the two outputs of the circuit that is, V_{out+} and V_{out-} are identical in common mode condition, so if we don't have any mismatch in the circuit, the differential output in common mode condition would be equal to 0. In fact, since g_{m1} and g_{m2} are equal and R_D has the same value in V_{out+} and V_{out-} sides, for the common mode gain we have:

$$A_{V_CM}(dB) = \left. \frac{R_D}{\frac{1}{g_{m1}} + 2 \times r_{o7}} \right|_{Left_side} - \left. \frac{R_D}{\frac{1}{g_{m2}} + 2 \times r_{o7}} \right|_{Right_side} = 0 \quad (4)$$

Where r_{o7} is the drain source resistor of M7. (the tail current source transistor which is connected to V_{bias} in fig.1). This shows that the required CMRR would be available without any specific effort and it does not make any change in our design procedure.

The other small signal parameters that we have to satisfy is the unity gain band width (ω_u) of the amplifier in deferential mode. According to the right half circuit of fig.1 in differential mode, the amplifier circuit shows two poles at node A (Drain of M_1) and output node (V_{out+}). The values of these two poles are as bellow:

$$\omega_{p1}(@node\ A) = \frac{1}{R_D \times C_A}$$

while

$$C_A \simeq C_{gs4} + C_{gd4} + C_{gd2} + C_{db2} \simeq C_{gs4} + C_{db2}$$

and

(5)

$$\omega_{p2}(@output\ node) \simeq \frac{g_{m4}}{C_{out}}$$

while

$$C_{out} \simeq C + C_{db3} + C_{gd3} + C_{sb4} \simeq C = 20\text{pF}$$

To determine the dominant pole, we have to consider the resistance and the capacitance which make these two poles. Although the resistance of node A is higher than output node, the output capacitance is much larger than the capacitance seen from node A. So considering the output capacitance of 20Pf, it seems that it is much easier to assume output pole as the dominant pole.

Now, to have the required $\omega_u = 2\pi \times 500\text{MHz}$, we have to pick up the size of M2 and M4 such that the summation of $C_{gs4} + C_{db2}$ doesn't exceed the borderline values of capacitor at node A. In fact to make sure that our assumption about dominant pole is correct we need to put the pole of node A in higher frequency than ω_u . In this case, we have:

$$\omega_{p1}(@node\ A) = \frac{1}{R_D \times C_A} > \omega_u$$

→

$$\frac{1}{500 \times C_A} > 2 \times \pi \times 500\text{MHz}$$

(6)

→

$$C_A < 1.5\text{PF}$$

→

$$C_{gs2} + C_{db1} < 1.5\text{PF}$$

According to the values of parasitic capacitors in 0.13um CMOS technology for usual sizing of M2 and M4 the parasitic capacitors would be less than 1.5PF which satisfy the above equation. But, it is necessary to check the parasitic capacitors after determination of M2 and M4 sizing.

Now it is easy to find the value of g_{m4} due to required ω_u . To find the g_{m4} value we can use the relation between Gain, ω_p dominant pole and ω_u . That is,

$$A_{v_diff}(s) = \frac{Gain_{DC}}{\left(1 + \frac{s}{\omega_{p1}}\right) \cdot \left(1 + \frac{s}{\omega_{p2}}\right)} = \frac{5}{\left(1 + \frac{s}{\omega_{p1}}\right) \cdot \left(1 + \frac{s}{\omega_{p2}}\right)}$$

for

$$\omega_u < \omega_{p1} \rightarrow \text{ignor } \omega_{p1} \rightarrow$$

$$A_{v_diff}(s) \approx Gain_{DC} \times \frac{1}{\left(1 + \frac{s}{\omega_{p2}}\right)} = \frac{5}{\left(1 + \frac{s}{\omega_{p2}}\right)}$$

$$\text{at } \omega = \omega_u \rightarrow \left|A_{v_diff}(s)\right|_{s=j\omega_u} = 1$$

$\rightarrow$

$$\left| \frac{5}{\left(1 + \frac{j\omega_u}{\omega_{p2}}\right)} \right| = 1 \tag{7}$$

$\rightarrow$

$$(5)^2 = 1 + \left(\frac{\omega_u}{\omega_{p2}}\right)^2$$

$$\omega_u = 5 \times \omega_{p2}$$

$\rightarrow$

$$2 \times \pi \times 500 \text{ MHz} = 5 \times \frac{g_{m4}}{C} = 5 \times \frac{g_{m4}}{20 \text{ PF}}$$

$\rightarrow$

$$g_{m4} = 12.5 \text{ mS}$$

$\rightarrow$

with additional margin

$$g_{m4} = 15 \text{ mS}$$

We have considered $g_{m4} = 15^{mS}$ to make sure that $\omega_u > 2\pi \times 500MHz$ is achievable after simulation of the circuit.

The third specification of the amplifier which should be satisfied by the designers is the swing at the amplifier output. According to the problem the amplifier should have 0.6V peak to peak voltage swing at the output in differential mode. So, for single ended swing in each side of the amplifier, we have:

$$\text{single_ended_swing} = \frac{600/2}{2} = \pm 150mV$$

That means each output voltages should be able to vary $\pm 150mV$ around its bias voltage without forcing any of the 3 existing transistors in the half circuit to enter the linear or cut-off regions. So it is important to set properly the output DC bias voltage. To find the proper DC bias voltage, first we should know the output voltage range that keeps all transistors in saturation region. This range can be calculated by determination of saturation region for each transistor as bellow:

$$M_3 \text{ at saturation region} \rightarrow V_{out,min} = V_{ds,sat\ M3} = V_{OD3} \quad (8)$$

The maximum voltage level is limited by M4 and the output bias voltage of the first stage of the amplifier that is the drain of M2. In fact, increase in output voltage can push the M4 transistor to linear or even cut-off region. Since the instant current of M4 is produced by the M7 bias current as the tail current source when M2 is on, it prevents M4 goes easily to the cut-off region and as a result, cut-off region for M4 is not critical here. From the circuit, it is seen that the limitation exposed by M4 when M2 entering to cut-off region and output voltage of the first stage goes up to VDD is much severe that linear region of M4 and it occurs sooner than the time M4 goes to linear region. Considering the cut-off condition of M2, we have:

$$M_2 \text{ cut-off} \rightarrow I_{M2} = 0 \rightarrow V_{out,1^{st} \text{ stage}} = V_{DD} \rightarrow \quad (9)$$

$$V_{out,max} = V_{DD} - I_{M2} \times R_D - V_{GS,M4} = V_{DD} - V_{GS,M4} = V_{DD} - V_{th,M4} - V_{ds,sat\ M4}$$

It is worth to consider that the value of $V_{th,M4}$ is larger than $V_{th0,M4}$ because of the big voltage difference between source to bulk voltage of M4 when M4 enters cut-off region and should be calculated from the V_{th} formula.

Based on above calculation in which the minimum and maximum possible voltages at the amplifier output have been derived, we expect that the maximum possible swing at the output of the amplifier to be:

$$V_{output_swing_single_ended_pp}(\text{max}) = V_{DD} - V_{GS,M4} - V_{ds,sat\ M3} = V_{DD} - V_{th,M4} - V_{ds,sat\ M4} - V_{ds,sat\ M3} \quad (10)$$

As a result, to have the maximum symmetrical voltage swing at the output, the best output DC voltage should be close to:

$$V_{output_DC_max_swing} = V_{ds,sat\ M_3} + \frac{V_{DD} - V_{th,M_4} - V_{ds,sat\ M_4} - V_{ds,sat\ M_3}}{2} \quad (11)$$

Now, using equation (1) to (11) and considering the parameters of the transistors which have been extracted from simulation of an NMOS transistor in 0.13um CMOS technology and are shown in Table.1 we would be able to determine the sizing of transistors and bias voltages of the circuit to satisfy all the requirements of the circuit.

Table.1. Transistor Parameters

Parameter	Value
$\mu_n \cdot C_{ox}$	$350 \frac{\mu A}{V^2}$
V_{th}	0.38 V
λ	$0.5 V^{-1}$
γ	0.45

To start calculation of the transistors' W and L based on the achieved values for gm we can consider a reasonable value for overdrive voltage of output transistors. We start with 0.1V to 0.2V voltage overdrive because it is far from the subthreshold region and also it is not too large so it would not limit the swing of the amplifier.

Considering $V_{ds,sat4} = V_{OD4} = 0.2$ for M4 and $V_{ds,sat3} = V_{OD3} = 0.1$ for M3 based on equation(11) for the optimum value of output DC voltage to have maximum symmetrical swing at the output, we have:

$$V_{output_DC_max_swing} = V_{ds,sat\ M_3} + \frac{V_{DD} - V_{th,M_4} - V_{ds,sat\ M_4} - V_{ds,sat\ M_3}}{2} \quad (12)$$

$$\rightarrow V_{output_DC_max_swing} = 0.1 + \frac{1.3 - 0.45 - 0.2 - 0.1}{2} = 0.325V$$

In equation (12) we have considered the value of $V_{th,M_4} = 0.45V$ instead of $0.38V$ since it has changed because of body effect. The value of $0.45V$ is achieved by solving equation(12) and the value of Vth from equation (13) together and also by doing a simple simulation to achieve it.

$$V_{th} = V_{th0} + \gamma \times (\sqrt{\Phi + V_{SB}} - \sqrt{\Phi}) \quad (13)$$

and

$$V_{SB} = V_{output_DC_max_swing}$$

It is worth mentioning that, by adjusting $V_{output_DC} = V_{output_DC_max_swing} = 0.325V$ the output achievable swing would be:

$$V_{out+ swing} = \frac{V_{output_swing_single_ended_pp}(\max)}{2} = \frac{V_{DD} - V_{th,M_4} - V_{ds,sat M_4} - V_{ds,sat M_3}}{2} = \pm 0.25V \quad (14)$$

This value is $\pm 100mV$ larger than the required swing at the output of amplifier. So we can reduce the $V_{out+ DC}$ from 0.325V to 0.25V that is 100mV which would relax the determination of the bias current of the first stage. In fact, by selecting $V_{out+ DC} = 0.25V$ we can increase bias current of the first stage, consequently we would have higher gm and it would be much easier to reach the required gain of the first stage. Now according to the value of $V_{out+ DC} = 0.25V$ we can calculate the current of I_{D2} as follow:

$$\begin{aligned} V_{out+ DC} &= V_{DD} - I_{D2} \times R - V_{GS4} = V_{DD} - I_{D2} \times R - V_{th4} - V_{ds,sat4} \\ &\rightarrow \\ 0.25 &= 1.3 - I_{D2} \times 500 - 0.4 - 0.2 \\ &\rightarrow \\ I_{D2} &= 900\mu A \end{aligned} \quad (15)$$

In this case, the DC voltage of the first stage output would be:

$$V_{output_DC_1^{st} stage} = V_{DD} - I_{D2} \times R = 1.3 - 900\mu A \times 500 = 0.85V \quad (16)$$

Now based on the value of $g_{m1} = 18^{mS}$ which is obtained from equation (1) we have:

$$g_m = \frac{2 \times I_D}{V_{OD}} \text{ or } g_m = k'_n \times \frac{W}{L} \times V_{OD}$$

So

$$g_{m2} = \frac{2 \times I_{D2}}{V_{OD2}} \rightarrow$$

$$18^{mS} = \frac{2 \times 0.9^{mA}}{V_{OD2}} \rightarrow V_{OD2} = 0.1V$$

and

$$g_{m2} = k'_n \times \frac{W_2}{L_2} \times V_{OD2} \rightarrow$$

$$18^{mS} = 350^{\mu A/V^2} \times \frac{W_2}{L_2} \times 0.1 \rightarrow$$

$$\frac{W_2}{L_2} = 515 \rightarrow \text{for } L_2 = L_{\min} = 0.12^{\mu m} \rightarrow W_2 = 62^{\mu m}$$

For transistor M4, since we considered $V_{OD4} = 0.2V$ and based on the calculated $g_{m4} = 15^{mS}$ from equation (7) we have:

$$g_{m4} = \frac{2 \times I_{D4}}{V_{OD4}} \rightarrow$$

$$15^{mS} = \frac{2 \times I_{D4}}{0.2} \rightarrow I_{D4} = 1.5^{mA}$$

and

$$g_{m4} = k'_n \times \frac{W_4}{L_4} \times V_{OD4} \rightarrow$$

$$15^{mS} = 350^{\mu A/V^2} \times \frac{W_4}{L_4} \times 0.2 \rightarrow$$

$$\frac{W_4}{L_4} = 214 \rightarrow \text{for } L_4 = L_{\min} = 0.12^{\mu m} \rightarrow W_4 = 25^{\mu m}$$

For the overdrive voltage of the M3 transistor we can consider 0.1V, for more swing capability

In this case, since the bias currents of the M3 and M4 are identical, for the $\frac{W_3}{L_3}$ we have:

$$I_{D4} = I_{D3} = 1.5^{mA}$$

and

$$V_{OD3} = \frac{1}{2} \times V_{OD4} = 0.1V \quad (19)$$

→

$$\frac{W_3}{L_3} = \sqrt{2} \times \frac{W_4}{L_4} = 300$$

It should be noted that for M3 it is better to pick larger length than the minimum L. That is because, the gm of M3 does not have any effect on the amplifier performance and only the ro of M3 is seen at the output node. So, by choosing larger L we can completely ignore the value of ro3 compare to gm of M4 at the output node. On the other hand, if we consider L3 too large it may change the output capacitance which is not desirable. Here we consider the value of L3 about Lmin for simplification. As a result, we have:

$$L_3 = L_{\min} = 0.12^{\mu m}$$

and

$$\frac{W_3}{L_3} = 300 \quad (20)$$

→

$$W_3 = 36^{\mu m}$$

According to the symmetry of the circuit the sizing of the symmetric transistor should be equal. That is, the size M1 and M2, M4 and M5 and M3 and M6 are identical and they have similar sizing for W and L.

Now it is time to calculate the W and L of M7. Due to differential structure of the circuit in fig.1 the bias current of M7 is equal to the summation of $I_{D1} + I_{D2} = 0.9 + 0.9 = 1.8^{mA}$. Considering 0.2V for V_{OD7} we have:

$$\begin{aligned}
I_{D7} &= \frac{1}{2} \times k_n' \times \frac{W_7}{L_7} \times V_{OD7}^2 \\
&\rightarrow \\
1.8^{mA} &= \frac{1}{2} \times 350^{\mu A/V^2} \times \frac{W_7}{L_7} \times (0.2)^2 \\
&\rightarrow \\
\frac{W_7}{L_7} &= 257
\end{aligned} \tag{21}$$

We should remember that, since we don't have tight limitation on the swing in the first stage, we can consider higher value for V_{OD7} to reduce $\frac{W_7}{L_7}$. In fact, the DC value of the drain voltage of M2 (the output of the first stage) is 0.85V based on (16). So, even with $\pm 0.2V$ single ended swing at this point we would have 0.65V headroom to provide the required DC bias voltage for V_{DS} of M2 and M7 to keep them in saturation region. Since we considered $V_{OD2} = 0.2V$ before, we can allocate up to 0.45V to V_{OD7} . For example we can set V_{OD7} at 0.35V. With $V_{OD7} = 0.35V$ the $\frac{W_7}{L_7}$ ratio would get smaller which leads to smaller M7 transistor area and less parasitic too.

In this case we have:

$$\begin{aligned}
I_{D7} &= \frac{1}{2} \times k_n' \times \frac{W_7}{L_7} \times V_{OD7}^2 \\
&\rightarrow \\
1.8^{mA} &= \frac{1}{2} \times 350^{\mu A/V^2} \times \frac{W_7}{L_7} \times (0.35)^2 \\
&\rightarrow \\
\frac{W_7}{L_7} &= 84
\end{aligned} \tag{22}$$

As it is seen in equation(22), this 150mV change in V_{OD7} decreases the $\frac{W_7}{L_7}$ about 4 times. Now we can calculate the W_7 based on the V_{OD7} and I_{D7} when we consider $L_7 = L_{\min}$ as bellow:

$$\frac{W_7}{L_7} = 84 \quad \text{and} \quad L_7 = L_{\min} = 0.12^{\mu m} \rightarrow W_7 = 10.08^{\mu m} \quad (23)$$

The last step is to derive the appropriate values for the bias voltages of the circuit based on bias currents, overdrive voltage of transistors and the transistors' parameters.

The required input DC voltage at the gate of M7 to set the current of M1 at the required $I_{D7} = 1.8^{mA}$ is:

$$V_{bias} = V_{th,7} + V_{OD7} = 0.38 + 0.35 = 0.72V \quad (24)$$

The other DC voltage which we should have calculate is input DC voltage which is gate bias voltage of M1 and M2. With the mentioned $V_{OD} = 0.2V$ for M1 and M2 to keep both M2 and M7 in saturation region we would have:

$$V_{DC_in} = V_{GS,2} + V_{OD7} = V_{th,2} + V_{OD2} + V_{OD7} = 0.5 + 0.1 + 0.35 = 0.95V \quad (25)$$

And at last we can determine the DC bias voltage needed to bias gate of M3 and M6. The bias voltage of M3, M6 is equal to:

$$V_{DC_bias_M_3} = V_{th,3} + V_{OD3} = 0.38 + 0.1 = 0.48V \quad (26)$$

Now based on calculated values for transistors W, L and bias voltages of the circuit we can design and simulate the circuit in the Cadence software.

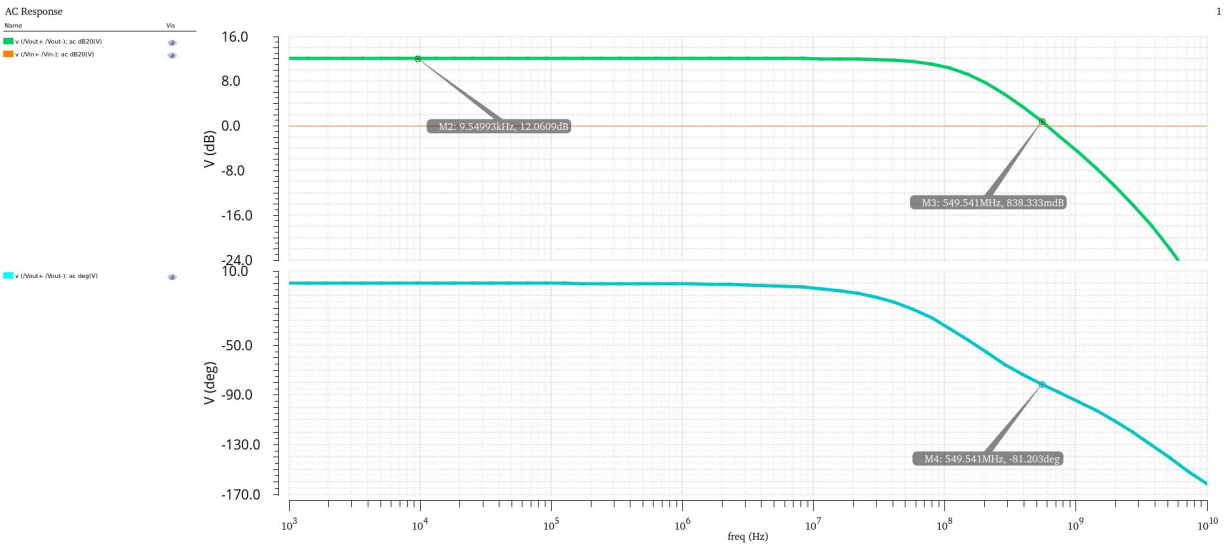
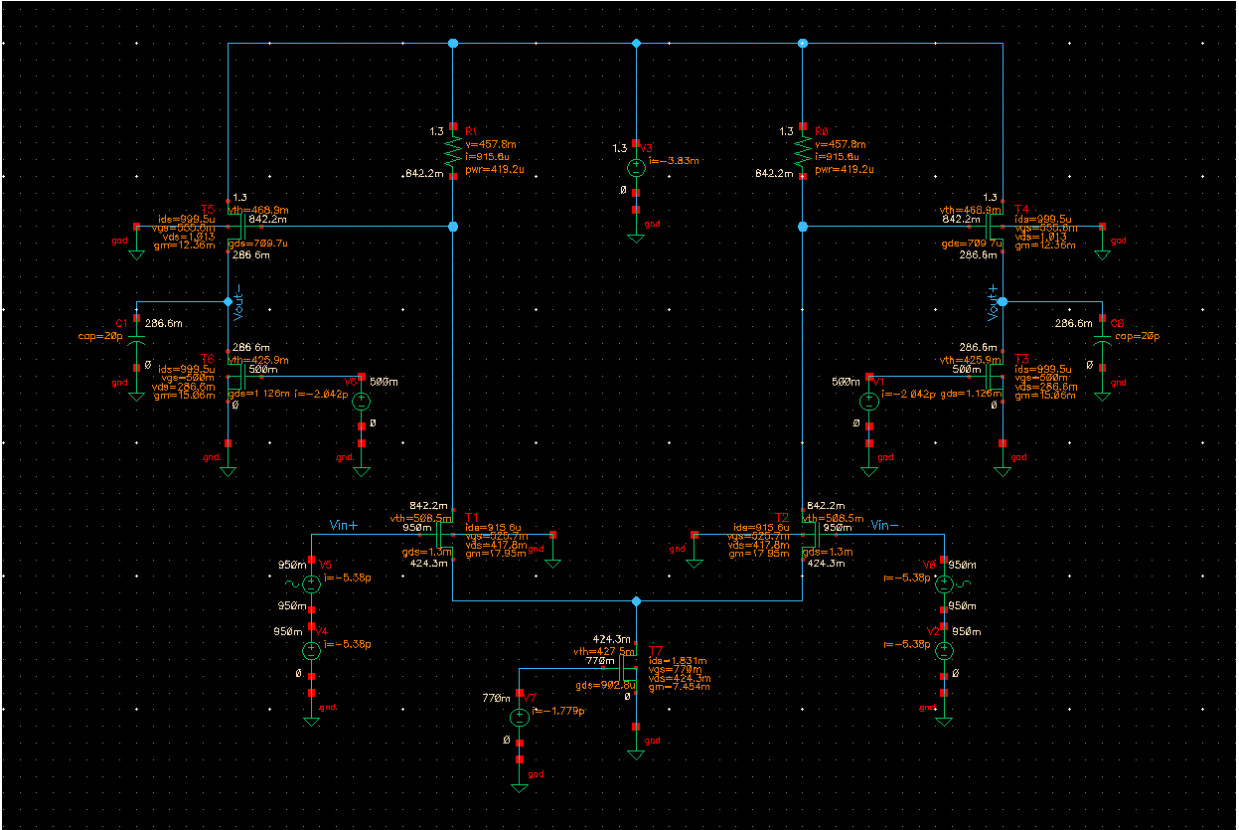
Simulation:

Based on the values achieved through calculation the circuit has been designed. Some modification in size and bias voltage has been done to achieve the required gm, gain and other specs of the circuit. In table 2 the final values for transistor sizing and bias voltage are shown.

Table.2. Real sizing after simulation

Parameter	Value
Current Consumption	3.8 <i>mA</i>
V _{in} (DC)	950 mV
V _{bias} (M7)	770mV
V _{bias} (M3 and M6)	500mV
$\frac{W_1}{L_1}$ and $\frac{W_2}{L_2}$	900
$\frac{W_4}{L_4}$ and $\frac{W_5}{L_5}$	250
$\frac{W_3}{L_3}$ and $\frac{W_6}{L_6}$	420
$\frac{W_7}{L_7}$	90

The circuit with DC operating points has been shown in fig.2. The simulation results for AC response and transient response are shown in fig.3 and fig.4 respectively. The CMRR is shown in fig.5. As it is seen, all the requirements of the circuit are in acceptable range.



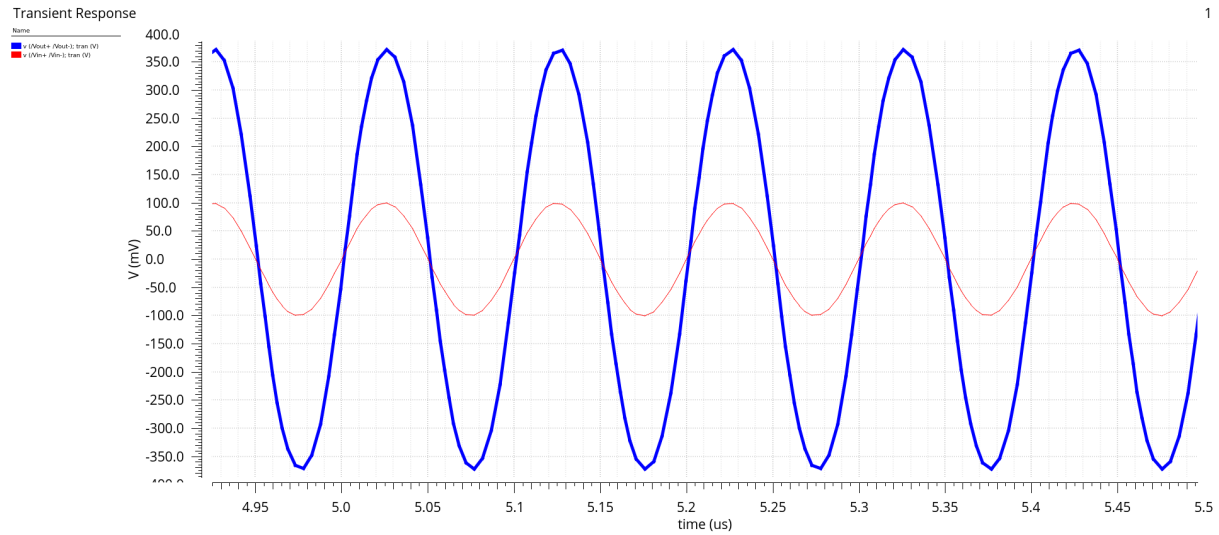


Fig.4. Transient Response. Output differential voltage (blue) and input differential voltage (red)

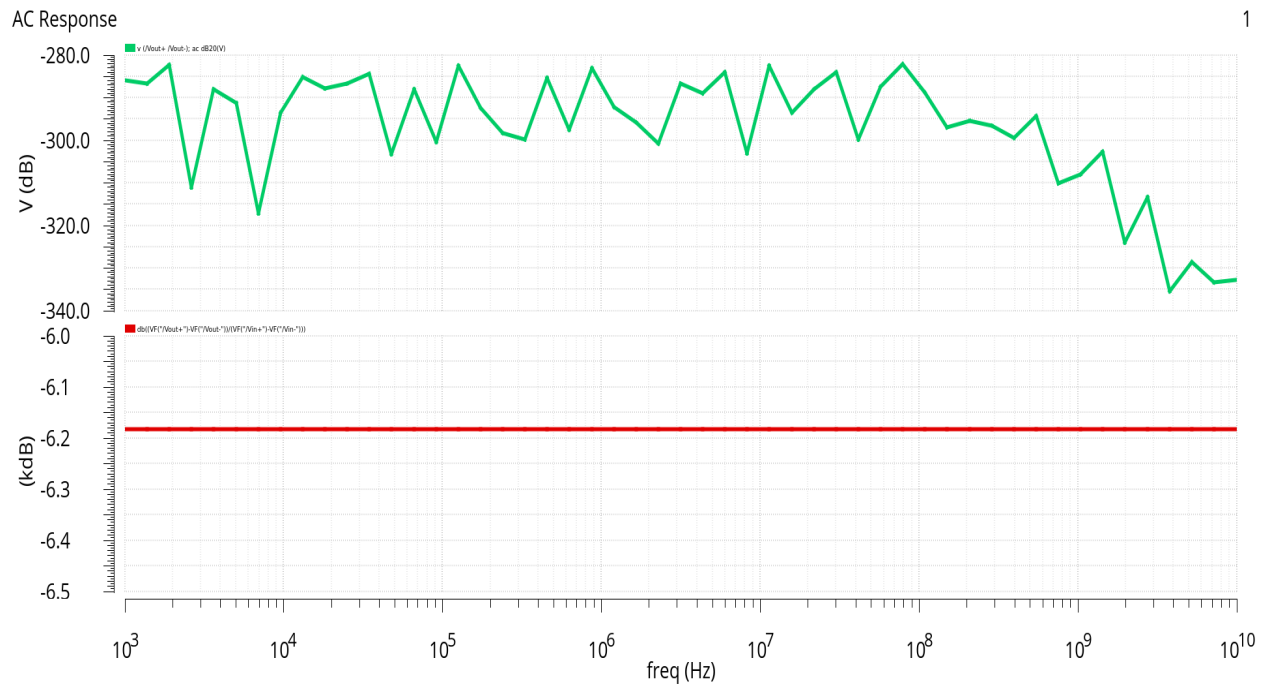


Fig.5. AC Response (CMRR)